

SCAN Disk Scheduling Algorithm/Elevator Algorithm-

- As the name suggests, this algorithm scans all the cylinders of the disk back and forth.
- Head starts from one end of the disk and move towards the other end servicing all the requests in between.
- After reaching the other end, head reverses its direction and move towards the starting end servicing all the requests in between.
- The same process repeats.

NOTE-

- SCAN Algorithm is also called as **Elevator Algorithm**.
- This is because its working resembles the working of an elevator.

Advantages-

- It is simple, easy to understand and implement.
- It does not lead to starvation.
- It provides low variance in response time and waiting time.

Disadvantages-

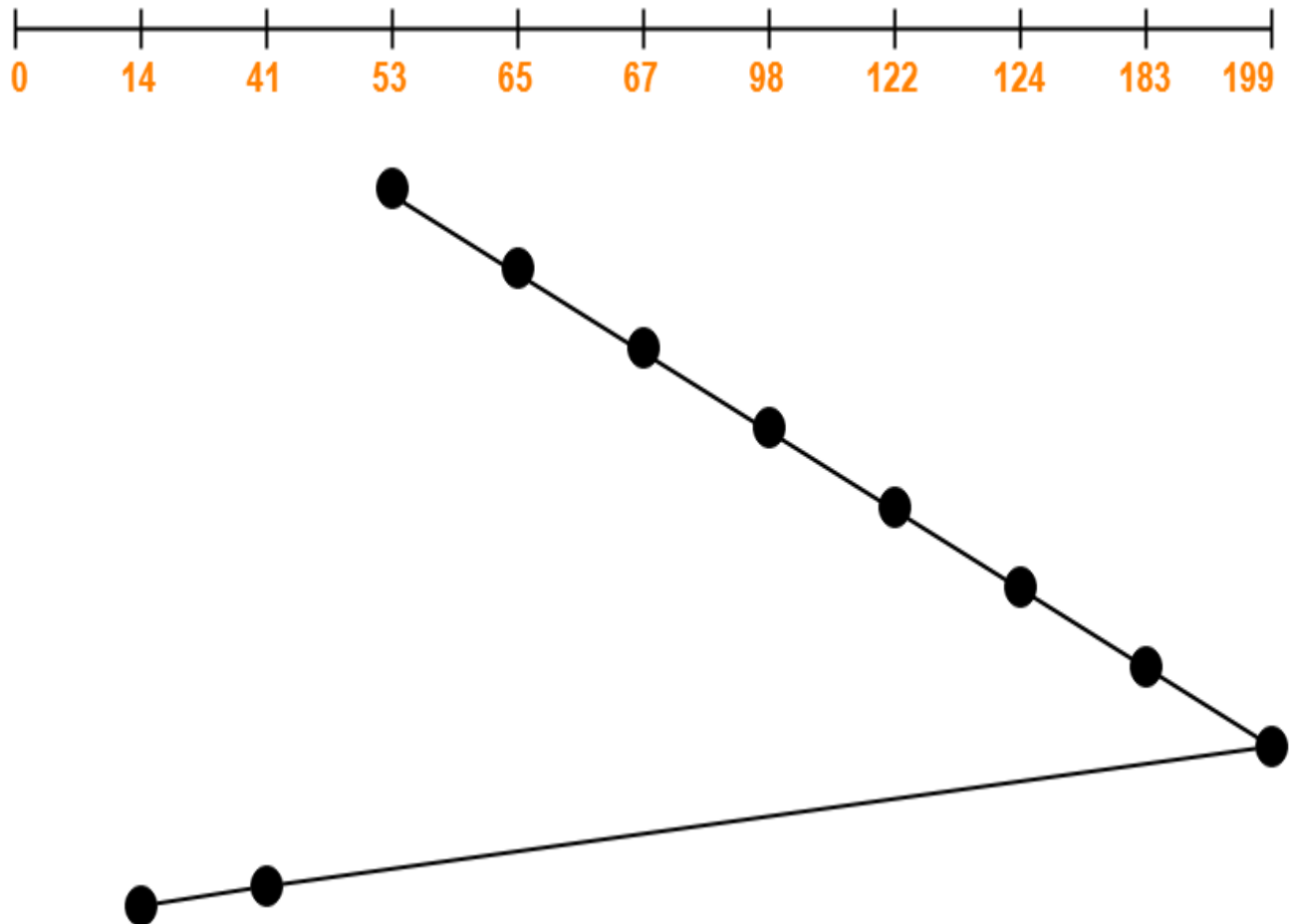
- It causes long waiting time for the cylinders just visited by the head.
- It causes the head to move till the end of the disk even if there are no requests to be serviced.

PRACTICE PROBLEM BASED ON SCAN DISK SCHEDULING ALGORITHM-

Problem-

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SCAN scheduling algorithm is used. The head is initially at cylinder number 53 moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____.

Solution-



Total head movements incurred while servicing these requests

$$= (65 - 53) + (67 - 65) + (98 - 67) + (122 - 98) + (124 - 122) + (183 - 124) + (199 - 183) + (199 - 41) + (41 - 14)$$

$$= 12 + 2 + 31 + 24 + 2 + 59 + 16 + 158 + 27$$

$$= 331$$

Alternatively,

Total head movements incurred while servicing these requests

$$= (199 - 53) + (199 - 14)$$

$$= 146 + 185$$

= 331